A FINAL YEAR PROJECT REPORT ON

DESIGN, FABRICATION, AND CORE LOSS MODELLING OF A HIGH-FREQUENCY TRANSFORMER FOR DUAL ACTIVE BRIDGE CONVERTER

Submitted in partial fulfillment of the requirements for the degree of
Bachelor of Technology in Electrical Engineering

by

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Abstract

This project report presents the complete design, hardware fabrication, and advanced core loss optimization of a 25 kW, 100 kHz High-Frequency Transformer (HFT) optimized for Dual Active Bridge (DAB) isolated DC-DC converters. In the first phase (Semester 7), the transformer's structural sizing, core geometry, and winding parameters were derived using the traditional Area Product method, resulting in the selection of a high-resistivity MnZn ferrite core (N87) and optimized interleaved Litz wire layouts to minimize high-frequency skin and proximity vulnerabilities. The second phase (Semester 8) expands this layout into detailed mathematical loss modelling and physical validation. Because traditional Steinmetz frameworks fail under non-sinusoidal voltage profiles typical of converter operation, the Improved Generalized Steinmetz Equation (iGSE) was implemented. Steinmetz material variables (α = 1.45, β = 2.8, k = 42.774 kW/m^3) were extracted from graphical loss trajectories, and a streamlined computational platform with a multi-waveform configuration toggle logic was engineered inside Excel. To connect theoretical calculations with real-world profiles, the physical transformer prototype was successfully completed. Finally, an integrated engineering verification loop is outlined for future work, utilizing high-fidelity Finite Element Analysis (FEA) simulations within ANSYS Maxwell to visually map magnetic flux behavior and confirm performance boundaries against experimental hardware metrics.

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1. Introduction & Problem Statement

Modern power electronic infrastructures demand ultra-high efficiency, bidirectionality, and high power density to satisfy the rigid operational constraints of electric vehicle (EV) fast-charging ports, microgrids, and renewable energy conversion links. Among various isolated DC-DC topologies, the Dual Active Bridge (DAB) converter stands out as a leading solution, providing symmetrical power flow control via phase-shift modulation techniques and inherent soft-switching capability (Zero Voltage Switching). A schematic representation of the standard DAB topology is modeled below, exhibiting the two full-bridge active stages coupled through a central isolation element.

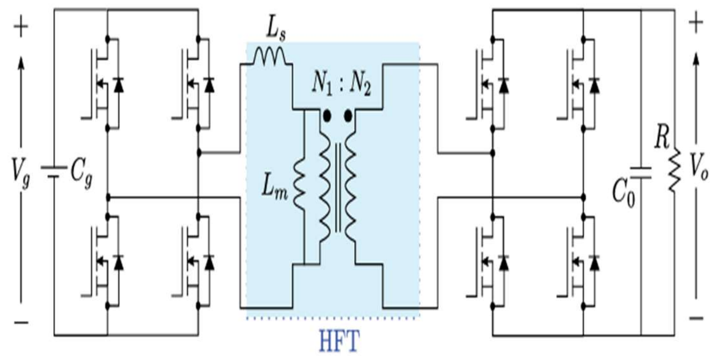


Fig. 1: Schematic of a Dual Active Bridge (DAB) converter

The critical technical bottleneck in a DAB system is the High-Frequency Transformer (HFT). The HFT must satisfy rigorous performance indices: transferring 25 kW of power at a switching frequency of 100 kHz while maintaining low thermal signatures and small volumetric profiles. The input voltage swings between 180 V and 270 V (nominal 240 V), while the output matches a high-voltage battery link of 300 V to 420 V (nominal 360 V). At these high operational frequencies, standard structural design rules are obsolete due to complex high-frequency magnetic phenomena, including excessive core hysteresis losses and frequency-dependent copper losses (skin and proximity effects). This project outlines a unified solution covering the original sizing and winding design (Semester 7), parameters extraction and advanced core loss prediction via the iGSE methodology, and successful physical fabrication (Semester 8).

2. Core Selection & Material Optimization

The core represents the foundation of the high-frequency transformer design, dictating the maximum operating flux density before saturation and the absolute base loss density. The core material comparison focuses heavily on Manganese-Zinc (MnZn) ferrites versus advanced iron-based nanocrystalline ribbons. While nanocrystalline alloys possess an ultra-high saturation flux density (approximately 1.2 T) and minimal baseline loss profiles at moderate frequencies, they are highly expensive, mechanically fragile, and less available in geometric profiles suited for manual winding optimization within India. Conversely, MnZn ferrites exhibit high electrical resistivity, eliminating bulk macro-eddy current losses, and remain highly cost-effective.

Ap= Ac*Aw

V=N dϕ /dt

$$K_w = \frac{N_1 A_1 + N_2 A_1}{A_w}$$

$$B_p = \frac{V_1 D}{4 f_s N_1 A_c}$$

$$A_p = \frac{VA}{2 \times f \times B_p \times K_w \times J}$$

L=(N2Φp)/(Ts/2)

V=4fBpAcN

Substituting the nominal design parameters—Power = 25,000 VA, switching frequency (f_s) = 100 kHz, conservative maximum flux density limit (B_max) = 0.25 T, standard window fill factor (K_w) = 0.3, and targeted current density (J) = 3 A/mm2—the required minimum Area Product equals **5.5 x 10^-7 m4** (or 55 cm4). Based on this outcome, a premium large-scale E-core structural arrangement (such as a double-stacked configuration or an **E80/38/20** frame) meeting an operational A_p threshold of >= 61 cm4 was successfully selected, providing a secure safety margin against localized thermal saturation.

3. High-Frequency Conductor Sizing:

Operating at 100 kHz completely alters the current distribution inside standard solid metal conductors. The AC current is forced to crowd near the outer boundary of the wire due to the skin effect, governed by the standard skin depth (δ) relationship:

$$Skin\ Depth = \delta = \sqrt{\frac{\rho}{\pi f \mu}} = \sqrt{\frac{\rho}{\pi f \mu_r \mu_o}}$$

Evaluating this for copper at 100 kHz yields a rigid skin depth of approximately 0.206 mm. Consequently, any solid wire with a radius exceeding this boundary will encounter massive high-frequency resistance inflation and catastrophic localized heating. Furthermore, the proximity effect caused by adjacent fields induces strong internal circulating eddy currents. To bypass these limitations, individually insulated Litz wire strands with a strict strand diameter of 0.3 mm (well within 2*δ) were chosen. The required cross-sectional areas were computed via the worst-case RMS currents. At nominal power, the low-voltage primary link carries an RMS current of 139 A, requiring 46 mm2 of active copper area. The high-voltage secondary link draws 84 A, demanding 28 mm2.

By dividing the required copper area by the individual cross-section of a 0.3 mm strand (0.0707 mm2), the primary winding structure was defined as an interleaved bundle composed of 650 synchronized strands. The secondary winding configuration was defined as a bundle of 396 strands. The actual turns count was calculated using Faraday's limit under maximum voltage stress (V_in,max = 270 V), resulting in N_p = 4 turns for the primary, and N_s = 6 turns for the secondary.

4. Core Loss Modelling & Mathematical Framework (iGSE)

A major problem was the inability of standard manufacturer data to predict core loss under practical converter waveforms. The traditional Steinmetz Equation[3] ($P_v = k * f^{\alpha} * B_m^{\beta}$) is fundamentally bound to sinusoidal currents. When a square wave voltage excitation is applied, the resulting core flux is triangular, meaning the rate of change of magnetic flux (dB/dt) remains uniform throughout each linear segment. To accurately capture these continuous transitions, the Improved Generalized Steinmetz Equation (iGSE) [3]was used:

$$P_v = \frac{1}{T}\int_0^T k_i \left|\frac{dB}{dt}\right|^{\alpha} |\Delta B|^{\beta-\alpha}\, dt$$

$$k_i = \frac{k}{(2\pi)^{\alpha-1}\int_0^{2\pi} |\cos\theta|^{\alpha}\, 2^{\beta-\alpha}\, d\theta}$$

The iGSE calibration constant, k_i, is derived analytically to enforce exact compliance with standard sinusoidal datasheet parameters, eliminating divergence via a mathematical relationship utilizing Gamma functions:

$$\mathbf{k_i = k / ((2*\pi)^{(\alpha - 1)} * 2^{(\beta - \alpha)} * I\_\theta)}$$

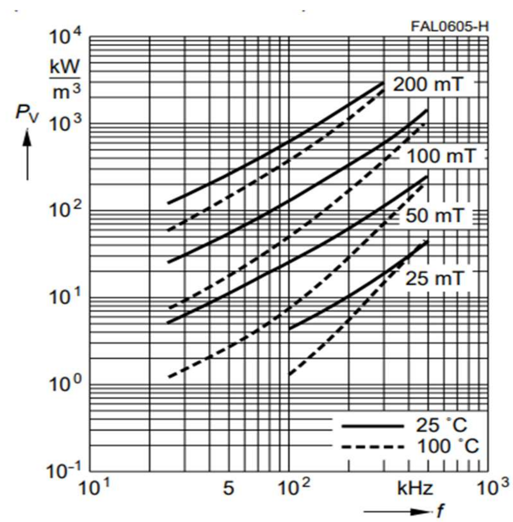
Where I_θ represents the definitive integral of |cos(θ)|^α from 0 to 2*π. For the TDK N87 soft ferrite material, the basic parameters were extracted in the next section by analyzing manufacturer log-log power loss plots over multiple operating points. The derived parameters are defined **as α = 1.45 and β = 2.8**. At the standard core reference **index (f = 100 kHz, B_m = 0.2 T, Temp = 100 degrees C)**, the baseline loss density equals 375 kW/m3. Substituting these variables into the classical framework gives a baseline datasheet coefficient of **k = 42.774 kW/m3.**

5. Extracting Steinmetz Parameters from Core Datasheets

To apply the iGSE core loss model for any selected magnetic core shape or material, the empirical Steinmetz parameters (α, β, and k) must be derived directly from the manufacturer's graphical datasheets. The extraction workflow is built upon isolating individual electrical dependencies log-log slope differentiation:

1. Calculation of Frequency Exponent (α):

Alpha represents the sensitivity of core loss with respect to frequency.



$\alpha = \partial \log(Pv) / \partial \log(f)$

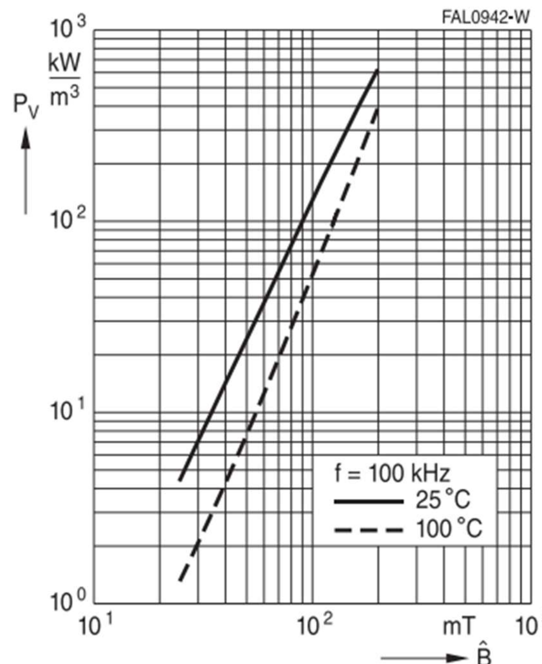
$\alpha = (\log(P2) - \log(P1)) / (\log(f2) - \log(f1))$

Using values obtained from the N87 ferrite datasheet

at B = 0.2T and temperature = 100°C, t

he calculated value is $\alpha \approx 1.45$

2. Calculation of Flux Density Exponent (β):
Beta represents the sensitivity of core loss with respect to flux density.



Graphs from[5]

$\beta = \partial \log(Pv) / \partial \log(B)$

$\beta = (\log(P2) - \log(P1)) / (\log(B2) - \log(B1))$

Using the datasheet values at

f = 100 kHz and temperature = 100°C, $\beta \approx 2.8$ 3.

Calculation of Steinmetz Coefficient (k):
After determining α and β, the Steinmetz coefficient k

can be calculated using a known datasheet operating point.

Given: Pv = 375 kW/m³

f = 100 kHz ,B = 0.2 T

α = 1.45, β = 2.8

Using the Steinmetz equation: $P_v = k f^{\alpha} B^{\beta}$

$375 = k \times (100000)^{1.45} \times (0.2)^{2.8}$

$k \approx 42.774$ kW/m³ (when frequency is expressed in kHz)

When frequency is expressed in Hz the constant becomes , $k_{eq} = 0.00191$ kW/m³

Thus the Steinmetz parameters for the selected operating condition are

$\alpha = 1.45$ $\beta = 2.8$ $k_{eq} = 0.00191$ kW/m³

6. Excel Automation Tool :

To automate these multi-variable calculations, an advanced core loss calculator was successfully designed in Excel. A critical problem identified during the development was that the raw datasheet coefficient 'k' assumes frequency is entered in kHz, whereas the iGSE mathematical integration must be computed using standard SI units (Hz and seconds). To bridge this scale gap without causing numerical errors, a frequency-equivalent correction factor was introduced directly into the cell hierarchy: $k_{eq} = k * (1000^{-\alpha})$.

iGSE Loss Tool: Multi-Waveform Support

SELECT WAVEFORM TYPE: Standard Square (Standard Square / Quasi-Square/PWM)

Parameter	Symbol	Value	Unit	Description
Steinmetz Alpha	alpha	1.45	-	Frequency exponent
Steinmetz Beta	beta	2.8	-	Flux density exponent
Steinmetz Coeff (k)	k	42.774	kW/m³	Datasheet k
Frequency	f	100000	Hz	Excitation frequency in Hz
Peak Flux Density	Bm	0.2	Tesla	Peak magnetic flux density
Duty Cycle (D)	D	0.5	-	Active pulse duty cycle (0-0.5)
Intermediate Calculations				
Equivalent k (keq)	keq	0.001910644	-	k adjusted for Hz
Integral (I)	I	3.538319753	-	iGSE Integral
Constant ki	ki	9.26427E-05	-	Derived iGSE Constant
delta B	deltaB	0.4	Tesla	Peak-to-peak flux
Core Loss Density (Pv)				
Loss Density	kW/m³	**345.9973899**		

Evaluating this for our parameters yields **k_{eq} = 42.774 * (1000^-1.45) = 0.00191 kW/m3.** This k_eq is subsequently routed to compute the core iGSE constant k_i. To maximize the utility of the tool for different converter operations, an interactive drop-down list cell was integrated, allowing immediate selection between two core operating modes .

To further validate the results, the findings were cross-referenced with established research (specifically Paper [8]). When operating at a frequency of **100 kHz** and a magnetic flux density of **0.2 T**, the core losses reported in the literature closely matched the values calculated by the **Excel-based analytical tool**.

6.1 Mathematical Derivation of dB/dt for Waveform Configurations:

To execute the iGSE integration, the explicit rate of change of flux density (dB/dt) must be established for each segment of the excitation period T_s.

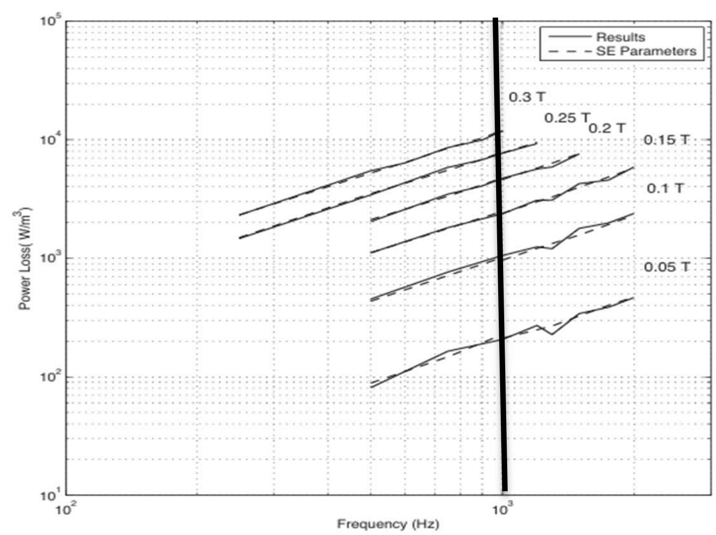


Fig 7: Measured core loss between 500 Hz and 2000 Hz and 0.05 T and 0.3 T with corresponding core loss predicted by Steinmetz Parameters

results from [8]

Derivation for Standard Square Wave Excitation (Two-Level Voltage):

Under a symmetric square-wave voltage, the flux density B(t) is a linear triangular waveform changing symmetrically between $-B_m$ and $+ B_m$. The total flux swing equals $\Delta B = 2* B_m$. The rising interval occurs during the on-time phase $D * T_s$ (where D = 0.5 for a standard square wave):

$$(dB/dt)\_rising = \Delta B / \Delta t = (2 * B_m) / (D * T_s) = (2 * B_m * f_s) / D$$

Similarly, the falling edge occurs during the remaining (1-D) * T_s segment of the operational window:

$|(dB/dt)\_falling| = \Delta B / \Delta t = (2 * B_m) / ((1 - D) * T_s)$

$= (2 * B_m * f_s) / (1 - D)$

Substituting these discrete uniform gradients into the iGSE core loss integral framework yields:

$$P_v = k_i(2B_m)^{\beta-\alpha}\left[\frac{(DT_s)\,|2B_m f_s D|^{\alpha} + ((1-D)T_s)\,|2B_m f_s 1 - D|^{\alpha}}{T_s}\right]$$

Factoring out the shared elements simplifies the final functional evaluation inside the Excel tool cell:

$$P_v = k_i(2B_m)^{\beta-\alpha}(2B_m f_s)^{\alpha}\left[D^{\,1-\alpha} + (1-D)^{\,1-\alpha}\right]$$

Derivation for Quasi-Square Wave Excitation (Three-Level PWM Voltage):

In quasi-square operations, the voltage waveform features an active positive window of width DT_s, followed by a zero-voltage dead time segment. A negative pulse of width D T_s follows, concluding with another zero-voltage interval. The flux remains flat at its plateau bounds during the zero blocks.

For the active rising and falling intervals, the gradient is defined as:

$|(dB/dt)\_active| = \Delta B / \Delta t = (2 * B_m) / (D * T_s) = (2 * B_m * f_s) / D$

During the inactive zero-voltage intervals, the core flux is completely constant, which forces the derivative to zero:

$(dB/dt)\_inactive = 0$

Integrating this piecewise profile across one full period T_s implies that only the two active regions contribute to core loss accumulation:

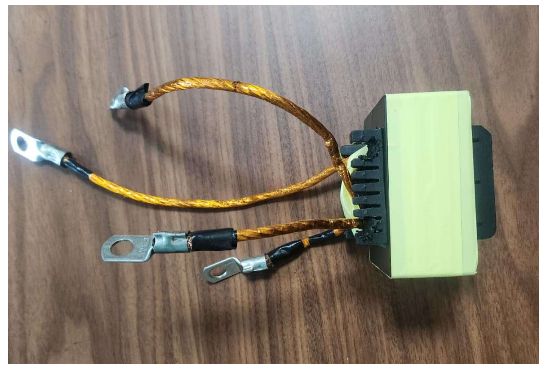
$$P_v = k_i\,(2B_m)^{\beta-\alpha}\left[\frac{2(DT_s)\,|2B_m f_s D|^{\alpha}}{T_s}\right]$$

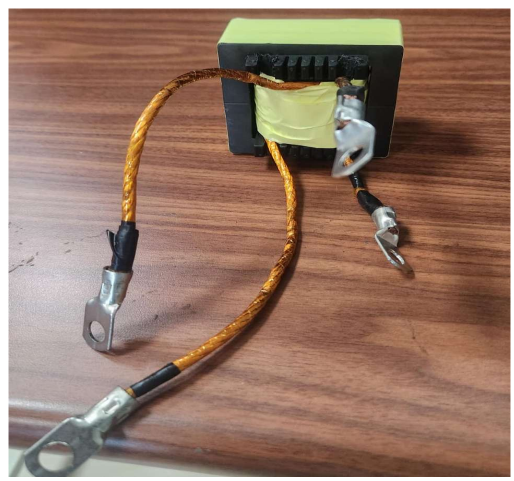
Simplifying the math yields the exact programmatic cell logic executed when the Quasi-Square toggle is engaged:

$$P_v = k_i\,(2B_m)^{\beta-\alpha}(2B_m f_s)^{\alpha}\left[2D^{\,1-\alpha}\right]$$

7. Hardware Fabrication and Assembly of HFT

To validate the analytical and mathematical models, a physical hardware prototype of the high-frequency transformer was fully assembled. The manufacturing stage required translating the theoretical parameters into a rugged mechanical layout. A high frequency transformer with the following specifications were designed and hardware was assembled.





Specifications:

P=5kw

Input Voltage Range=180-270V (nominal =240V)

Output Voltage Range =300-420V (nominal =360V)

Switching Frequency =100kHz

Assumed:

Current Density , $J_m = 2.5 \frac{A}{(mm)^2}$

Operating Flux Density , $B_m = 0.2T$

Window Utilization Factor , $K_u = 0.4$

Using area product method , $\mathbf{A_p = VA / (2 \times f \times Bp \times Ku \times J)}$

$A_p = 125000\ mm^4$

EE65/32/27 is being selected which has area product

Ap = $328490 mm^4$ ($A_c = 535\ mm^2$ $A_w = 614\ mm^2$)

For EE65 ,$A_c = 535 mm^2$

$N_p = \frac{V_p}{K_f * F_s * B_{pk} * A_c}$

$N_p = \frac{240*10^6}{4*100*10^3*0.2*535}$

$N_p = 6\ T$

- $N_s = \frac{V_s}{K_f * F_s * B_{pk} * A_c}$
- $N_s = \frac{360*10^6}{4*100*10^3*0.2*535}$
- $N_s = 9T$

Skin depth = $\delta = \frac{1}{\sqrt{\pi * F_{sw} * \mu * \sigma}}$ =0.213mm $(\sigma_{cu} = 5.8 * 10^7 \frac{s}{m})$

The radius of the selected strand should be less than δ

Conductor Cross Section Area Primary

- $A_{cu} = \frac{I_p}{J} = \frac{25}{2.5} = 10mm^2$

Primary winding strand Calculation

- N=$\frac{Total\ Area}{Area\ of\ each\ Conductor}$
- N=$\frac{10mm^2}{0.00785mm^2}$=1250 strands

Conductor Cross Section Area Secondary

- $A_{cu} = \frac{I_s}{J} = \frac{17}{2.5} = 6.8mm^2$

Secondary winding strand Calculation

- N=$\frac{Total\ Area}{Area\ of\ each\ Conductor}$
- N=$\frac{6.8mm^2}{0.00785mm^2}$=860 strands

Litz wire of 6mm^2 was selected for windings for primary paralleling was done as it required 10mm^2 cross section area

A **EE65/32/27** ferrite bobbin structure was secured, and high-purity braided Litz wire bundles were wound. To ensure strict adherence to the required low leakage profile, the primary turns (6 turns) and secondary turns (9 turns) were wrapped using a ultra-thin insulation sheets to prevent voltage breakdown.

The physical core halves were carefully aligned and minimize structural air-gap variances, which would otherwise corrupt the target magnetizing inductance. Testing is conducted using a high-frequency open-circuit and short-circuit operational loop powered by a flexible full-bridge inverter stage. The core loss is isolated by evaluating the open-circuit excitation power under high-frequency square wave voltage, matching the conditions modeled in the Excel tool.

8. Conclusion and Future Work

The high-frequency transformer design was mathematically established using the Area Product technique, and copper losses were mitigated via structured Litz bundles. The non-sinusoidal excitation losses were resolved by implementing the iGSE mathematical model within an automated Excel calculator, correctly utilizing a frequency-equivalent variable correction link.

Future Work Objective:

By running the transformer in an "open-circuit" state (where no load is attached), we can measure exactly how much power the magnetic core itself consumes. Using a **high-frequency square wave** that matches our earlier math models, we can isolate and measure the **core loss** of the transformer.

To verify the design's accuracy, the next phase involves building a high-fidelity 3D simulation of the transformer using **ANSYS Maxwell**. By inputting the exact physical dimensions and real-world magnetic properties of the **N87 ferrite material**, the simulation will mimic how the core behaves under actual operating conditions. The model will be tested with a **100 kHz square wave**, identical to the stress it faces in the real converter circuit. This allows us to "see" magnetic hot spots and energy losses throughout the core. Finally, these simulation results will be compared against our manual calculations and physical laboratory tests to ensure the design is fully validated from every angle..

9. References & Bibliography

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